

Academic Year: 2026- 2027

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Programme: B.Tech

Division: B

24SJPCST503 - MACHINE LEARNING

Course Project Guidelines

Introduction

Machine Learning is not simply about building models and obtaining predictions. A strong Machine Learning practitioner must be able to **ask the right questions, design appropriate experiments, interpret results, and explain why a model behaves the way it does.**

As part of the Continuous Internal Evaluation of the Machine Learning course, you will undertake a **small-scale Course Project – Investigation Challenge**. Unlike a conventional assignment, this activity is designed as an investigation. Each team will be given a specific Machine Learning question and will be expected to explore it through experimentation and evidence.

This course project is **not intended to be a complete software or application development project**, nor is it a replacement for the Machine Learning Laboratory. The laboratory provides opportunities to implement and practise Machine Learning techniques, whereas this activity focuses on the **deeper understanding behind those techniques.**

You will be encouraged to use appropriate datasets, tools, programming libraries, and AI-assisted tools where necessary. However, the objective is not to produce the most sophisticated code or achieve the highest accuracy. Instead, you should be able to demonstrate a clear chain of reasoning:

Question → Hypothesis → Experiment → Evidence → Analysis → Conclusion

Throughout the investigation, you should continuously ask:

What happened? Why did it happen? What evidence supports our explanation?

The final outcome should therefore demonstrate not only that you can *use* a Machine Learning technique, but that **you can think critically about its behaviour and limitations.**

Most importantly, every team member should understand the complete investigation and be able to defend the team's findings during the individual viva.

The guiding principle of this course project is: “Don’t just build a model. Find something out.”

1. Purpose of the Course Project

As part of the Continuous Internal Evaluation of the **Machine Learning** course, students are required to undertake a **small-scale Machine Learning Investigation.**

This is **not a complete software project** and is **not a replacement for the Machine Learning Laboratory.**

The objective is to develop your ability to:

- formulate a meaningful Machine Learning question;

- connect theoretical concepts with practical evidence;
- design an appropriate investigation;
- analyse experimental results;
- explain *why* results occur;
- critically evaluate limitations; and
- communicate and defend your conclusions.

The focus is:

Question → Hypothesis → Experiment → Evidence → Analysis → Conclusion

The highest marks will not necessarily go to the team with the highest accuracy or the most complicated code.

2. Important: This Is Different from the ML Lab

The Machine Learning Laboratory already provides opportunities to implement and experiment with ML algorithms.

Therefore, **do not treat this as another ML Lab experiment.**

ML Lab

"Can you implement and execute an ML technique?"

ML Course Project

"Can you investigate an ML question and explain what the evidence tells you?"

Your project should therefore focus on **reasoning, analysis and interpretation.**

3. Team Formation

The class has been divided into **18 teams of 4 students.**

The teams have been formed randomly while maintaining a reasonable gender distribution.

Important

- Every student must participate actively.
 - Every student must understand the **complete project**, not just their assigned portion.
 - During the individual viva, any student may be asked about any part of the project.
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4. Team-wise Investigation Challenges

Each team has been assigned a different investigation challenge.

Team	Investigation Topic	Research Question
Team 01	Model Complexity	When does increasing model complexity stop improving generalization?
Team 02	Class Imbalance	Can a classifier with high accuracy still be practically poor?
Team 03	Decision Threshold	How does the classification threshold change model behaviour?
Team 04	Feature Selection	Does adding more features necessarily improve prediction?
Team 05	Training Data	Does more training data always improve ML performance?
Team 06	Regularization	Can stronger regularization improve generalization while reducing training performance?
Team 07	Feature Scaling	Why does feature scaling affect some algorithms more than others?
Team 08	Hyperparameters	How can a hyperparameter change model behaviour?
Team 09	Model Interpretability	Is the highest-performing model always the best model?
Team 10	Evaluation Reliability	How reliable is a single train–test split as a performance estimate?
Team 11	Data Leakage	How can data leakage create misleadingly good ML results?
Team 12	Noise	How does noisy data affect ML generalization?
Team 13	Dimensionality	Does reducing dimensionality necessarily reduce predictive performance?
Team 14	Bias–Variance	How can experimental evidence reveal high bias and high variance?
Team 15	Model Comparison	When should a simpler model be preferred to a more complex model?
Team 16	Metric Choice	How does the choice of evaluation metric change the conclusion about a model?
Team 17	Data Quality	How can data quality influence the reliability of an ML conclusion?
Team 18	Generalization	Does a model that performs well on one dataset necessarily generalize to another?

5. Investigation Requirements for Each Team

Team 01 – Model Complexity

Research Question

When does increasing model complexity stop improving generalization?

Investigation

Investigate the relationship between model complexity and:

- training performance;
- validation/test performance;
- underfitting;
- overfitting; and
- generalization.

Minimum Requirement

Use at least **four different complexity levels**.

You must identify and explain the point at which increasing complexity becomes harmful or provides diminishing benefit.

Team 02 – Class Imbalance

Research Question

Can a classifier with high accuracy still be practically poor?

Investigation

Use an imbalanced classification dataset and investigate:

- accuracy;
- precision;
- recall;
- F1-score;

- confusion matrix.

Minimum Requirement

Demonstrate a situation in which accuracy alone could lead to a misleading conclusion.

Team 03 – Decision Threshold

Research Question

How does the classification threshold change model behaviour?

Investigation

Study how changing the classification threshold affects:

- predictions;
- false positives;
- false negatives;
- precision;
- recall;
- F1-score.

Minimum Requirement

Investigate at least **four different threshold values** and explain the observed trade-off.

Team 04 – Feature Selection

Research Question

Does adding more features necessarily improve prediction?

Investigation

Compare:

- a meaningful feature set;
- additional irrelevant features;
- redundant features, where appropriate.

Analyse the effect on model behaviour and generalization.

Team 05 – Training Data

Research Question

Does more training data always improve ML performance?

Investigation

Investigate model performance using different training-set sizes.

Minimum Requirement

Use at least **four training-data sizes**.

Discuss:

- improvement;
 - diminishing returns;
 - bias;
 - variance;
 - data quality.
-

Team 06 – Regularization

Research Question

Can stronger regularization improve generalization while reducing training performance?

Investigation

Study different regularization strengths and analyse:

- training performance;
- validation/test performance;
- model coefficients;
- model complexity.

Explain why regularization changes model behaviour.

Team 07 – Feature Scaling

Research Question

Why does feature scaling affect some algorithms more than others?

Investigation

Compare at least two suitable ML algorithms:

- with scaling;
- without scaling.

Important

Do not merely report which model performed better.

Explain the **theoretical reason** why scaling affects the algorithms differently.

Team 08 – Hyperparameters

Research Question

How can a hyperparameter change model behaviour?

Investigation

Select one suitable ML algorithm and one meaningful hyperparameter.

Investigate several values and analyse:

- model behaviour;
 - performance;
 - complexity;
 - overfitting/underfitting.
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Team 09 – Model Interpretability

Research Question

Is the highest-performing model always the best model?

Compare models considering:

- predictive performance;
- interpretability;
- complexity;
- computational requirements;
- suitability for the application.

Key Question

If Model A is slightly more accurate but much less interpretable, would you always select Model A?

Justify your answer.

Team 10 – Evaluation Reliability

Research Question

How reliable is a single train–test split as a performance estimate?

Investigate the variability of performance under different data splits and compare this with an appropriate resampling/cross-validation approach.

Discuss:

- reliability;
 - variation;
 - computational cost;
 - generalization.
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Team 11 – Data Leakage

Research Question

How can data leakage create misleadingly good ML results?

Identify and demonstrate one realistic example of data leakage.

Compare:

Correct methodology

with

Leakage-prone methodology

Explain why the results differ.

Team 12 – Noise

Research Question

How does noisy data affect ML generalization?

Introduce controlled noise into a suitable dataset and investigate its effect on model performance.

Explain:

- what changed;
 - why it changed;
 - how noise affects learning;
 - limitations of the experiment.
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Team 13 – Dimensionality

Research Question

Does reducing dimensionality necessarily reduce predictive performance?

Investigate the effect of reducing the number of features.

Discuss the trade-off between:

- dimensionality;
 - information;
 - computational cost;
 - performance.
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Team 14 – Bias–Variance

Research Question

How can experimental evidence reveal high bias and high variance?

Design an investigation that provides evidence of:

- high bias;
- high variance.

Explain the observations using the bias–variance framework.

Team 15 – Model Comparison

Research Question

When should a simpler model be preferred to a more complex model?

Compare suitable models considering:

- performance;
- complexity;
- interpretability;
- computational requirements;
- generalization.

Your conclusion should not simply be:

"Model X has higher accuracy, therefore Model X is better."

Team 16 – Metric Choice

Research Question

How does the choice of evaluation metric change the conclusion about a model?

Evaluate the same model/predictions using multiple appropriate evaluation metrics.

Explain why different metrics may lead to different conclusions.

Team 17 – Data Quality

Research Question

How can data quality influence the reliability of an ML conclusion?

Investigate one data-quality issue, such as:

- missing values;
- outliers;
- inconsistent values;
- noisy observations.

Show how the issue affects the investigation.

Team 18 – Generalization

Research Question

Does a model that performs well on one dataset necessarily generalize to another?

Use two related datasets or introduce a meaningful distribution shift.

Investigate whether the conclusions obtained from one data distribution remain valid under another.

6. What Every Team Must Do

Every team must follow these steps.

Step 1 – Research Question

Clearly state:

What are we trying to find out?

Step 2 – Hypothesis

State:

What do we expect to happen, and why?

Step 3 – Theory

Identify the ML concepts that explain the expected behaviour.

Step 4 – Experimental Design

Explain:

- what will be changed;
- what will be measured;
- what will remain controlled;
- what will be compared.

Step 5 – Evidence

Generate appropriate:

- numerical results;
- tables;
- graphs;
- visualizations.

Step 6 – Analysis

Answer:

Why did this happen?

This is the most important part of the project.

Step 7 – Critical Evaluation

Discuss:

- assumptions;
- limitations;
- reliability;
- possible sources of error.

Step 8 – Conclusion

Answer the original research question using evidence.

7. Minimum Evidence Requirement

Unless your topic requires something different, your investigation should include:

- At least **3 meaningful experimental conditions/comparisons**
 - At least **2 meaningful graphs or tables**
 - Quantitative evidence wherever appropriate
 - Interpretation of the results
 - Connection to ML theory
 - At least **two limitations**
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8. Dataset Selection

Teams may use an appropriate public dataset.

However, **dataset selection must be justified.**

Your report should explain:

- What is the dataset?
- What is the target variable?
- What are the important features?
- Why is this dataset suitable for your investigation?
- Are there any limitations or biases in the dataset?

Do not select a dataset simply because an AI tool recommended it.

9. Use of AI Tools

AI tools are permitted.

You may use:

- ChatGPT
- Gemini
- Claude
- GitHub Copilot

- AI coding agents
- other appropriate AI-assisted tools

for:

- understanding concepts;
- generating code;
- debugging;
- suggesting experiments;
- visualization;
- documentation;
- language improvement.

However:

AI is an assistant, not the investigator.

You are responsible for:

- checking AI-generated information;
- testing generated code;
- understanding the methodology;
- verifying results;
- correcting errors;
- defending your conclusions.

Important

During the viva, you may be asked:

"Why did you use this method?"

"What would happen if we changed this parameter?"

"Explain this graph."

"Why does this result occur?"

If you cannot explain your own project, AI-generated output will not help you earn marks.



10. AI Usage Declaration

Every team must include the following in the report:

AI Usage Declaration

AI tool(s) used:

.....

Purpose of using AI:

.....

Parts of the project assisted by AI:

.....

How the AI-generated output was verified:

.....

One AI suggestion that we modified, rejected, or corrected:

.....

Honest disclosure of AI usage is expected.

11. Report Format

The final report should contain:

1. Title

2. Team Members

3. Research Question

4. Hypothesis

5. Theoretical Background

6. Dataset / Data Source

7. Experimental Design

8. Results

9. Analysis and Interpretation

10. Critical Evaluation and Limitations

11. Conclusion

12. AI Usage Declaration

13. References

Recommended length

4–5 pages, excluding references.

Avoid:

- lengthy code listings;
- unnecessary screenshots;
- copied theoretical material.

Focus on **your investigation and findings**.

12. Presentation

Each team will give a presentation of approximately **5 minutes**.

Suggested 7-slide structure

Slide 1: Title & Team Members

Slide 2: Research Question & Hypothesis

Slide 3: Relevant ML Theory

Slide 4: Experimental Design

Slide 5: Key Results

Slide 6: Analysis – Why did this happen?

Slide 7: Conclusion, Limitations & AI Reflection

Remember

The presentation should answer:

What did we investigate?

What did we find?

Why did we find it?

13. Individual Viva / Defence

Each student will have an **individual viva**.

The viva is worth **3 marks**.

Questions will not be limited to memorized definitions.

You may be asked:

Understanding

- What exactly did your team investigate?
- What was your hypothesis?
- What ML concept is central to your investigation?

Interpretation

- What does this graph show?
- Why did the performance change?
- Why is this result important?

Reasoning

- Why did you choose this dataset?
- Why did you choose this metric?
- Why did you use these experimental conditions?

What-if Questions

The faculty member may change a condition and ask you to predict what would happen.

For example:

"What if the training data were reduced?"

"What if the dataset became highly imbalanced?"

"What if λ were increased substantially?"

"What if the decision threshold were changed?"

"What if an important feature were removed?"

You should be able to reason about the answer using ML concepts.

14. Assessment – 15 Marks

Component	Marks
Research Question, Hypothesis & Conceptual Background	2
Experimental Design	3
Results & Evidence	3
Critical Analysis & ML Reasoning	4
Individual Viva / Defence	3
TOTAL	15

What receives the highest marks?

The most important component is:

Critical Analysis & ML Reasoning – 4 Marks

You should be able to explain:

WHAT happened?

and more importantly:

WHY did it happen?

15. Three-Week Schedule

WEEK 1 – DEFINE

- Understand the assigned challenge
- Select dataset
- Formulate hypothesis
- Design experiment
- Prepare one-page proposal

Deliverable:

Project Proposal

WEEK 2 – INVESTIGATE

- Conduct experiments
 - Generate evidence
 - Prepare graphs/tables
 - Analyse preliminary results
 - Review the experimental design
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WEEK 3 – ANALYSE & DEFEND

- Complete analysis
 - Prepare final report
 - Prepare presentation
 - Submit notebook/supporting evidence
 - Individual viva
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16. Important Rules

1. The assigned investigation topic must be followed.
2. Do not simply reproduce an ML Lab experiment.
3. Every team member must understand the complete project.
4. Do not fabricate or manipulate results.
5. If your results contradict your hypothesis, **report and explain them honestly**.
6. All datasets and external sources must be acknowledged.
7. AI tools are permitted, but their use must be declared.
8. Students are responsible for verifying AI-generated content.

9. The notebook/code is supporting evidence; code complexity is not the primary assessment criterion.
10. Marks are based on **understanding, reasoning, evidence and analysis**.
11. The faculty member may ask any student about any part of the project.

17. Final Guiding Principle

DON'T JUST BUILD A MODEL. FIND SOMETHING OUT.

A successful project is not necessarily the one with:

- ✗ the highest accuracy
- ✗ the biggest dataset
- ✗ the longest code
- ✗ the most complicated algorithm
- ✗ the most attractive interface

A successful project has:

A clear question



A logical experiment



Reliable evidence



Correct ML reasoning



Critical analysis



A meaningful conclusion

Your final conclusion should answer three questions:

What happened?

Why did it happen?

How confidently can we make this conclusion?



Sarju S